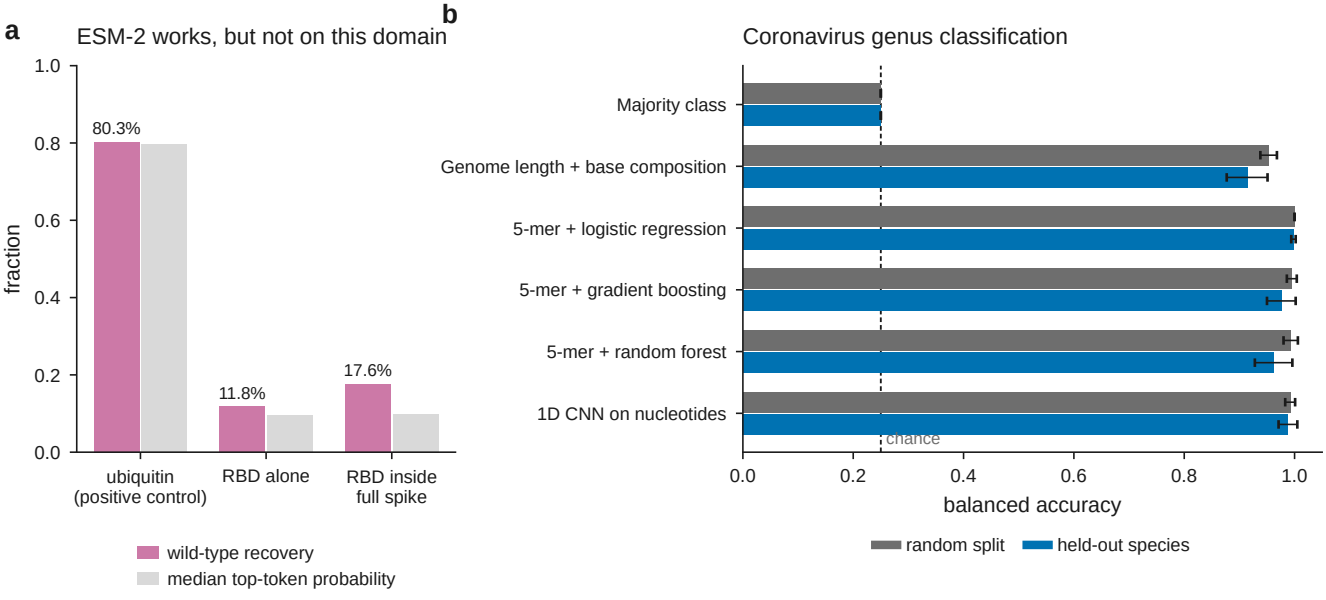
a
ESM-2 works, but not on this domain
fraction
1.0
0.8
0.6
0.4
0.2
0.0
80.3%
11.8%
17.6%
ubiquitin
(positive control)
RBD alone
RBD inside
full spike
wild-type recovery
median top-token probability
b
Coronavirus genus classification
Majority class
Genome length + base composition
5-mer + logistic regression
5-mer + gradient boosting
5-mer + random forest
1D CNN on nucleotides
chance
0.0
0.2
0.4
0.6
0.8
1.0
balanced accuracy
random split
held-out species